

Figure 1. Physiology-Informed Time-Aware Causal Transformer (PITACT) Architecture.

Raw ICU Time Series $x(t)$
Values $v(t)$ | Masks $m(t)$ | Deltas $dt(t)$
[34 Features x 3 = 102 Dimensions]

Strict Causal Transformer Encoder
Upper-Triangular Mask ($t' \leq t$ Only)
Zero Future Information Leakage

Causal Novelty Feature Layer
1. Reliability Decay $R(t) = \exp(-\gamma * dt)$
2. Dynamics: Velocity v' & Acceleration v''
3. Patient Baseline Deviation $v_{dev}(t)$

Dynamic Organ Interaction Node
Cardio, Resp, Renal, Liver, Metab, Temp
 $A_{ij}(t)$ Dynamic Attention Coupling

Multi-Horizon Output Heads
Primary 6h Head | Secondary 12h & 24h Heads
Lead-Time-Aware Training Loss

External Validation on Emory (Set B)
AUROC = 0.9715 | AUPRC = 0.4560
Official Utility = +0.6915
100% Causal Invariance Passed